WORKED SOLUTION

STEP II Specimen paper, Question 5

Pure • Series and Integral Bounds

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q5

The key idea

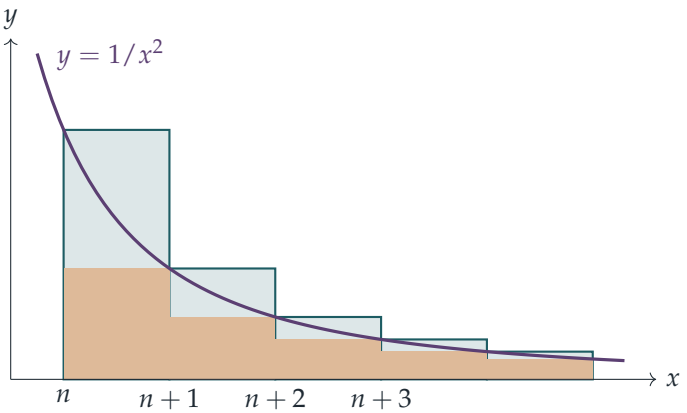
A sum of values of a decreasing function can be pictured as a row of unit-width rectangles, and the integral as the area under the curve. Putting the rectangles to the right of each sample point makes them stick out above the curve; putting them to the left tucks them underneath. That gives the sandwich, and everything after it is bookkeeping: the tail of the series is trapped between two simple fractions, so adding a guess for the tail improves the estimate enormously.

Solution

The sandwich

Let $f(x) = 1/x^2$, which is positive and strictly decreasing for $x > 0$. We also need

$$\int_n^\infty \frac{\mathrm{d}x}{x^2} = \lim_{M\to\infty}\left[-\frac{1}{x}\right]_n^M = \frac{1}{n}.$$



Upper rectangles (teal outlines). On $[r, r+1]$ put a rectangle of height $f(r) = 1/r^2$. Since

f is decreasing, $f(x) < f(r)$ for $r < x \leq r+1$, so this rectangle has area $1/r^2$ strictly greater than $\int_r^{r+1} f(x)\,\mathrm{d}x$. Adding these for $r = n, n+1, \ldots$ gives

$$\sum_{r=n}^{\infty} \frac{1}{r^2} > \int_n^{\infty} \frac{\mathrm{d}x}{x^2}.$$

Lower rectangles (amber). On $[r-1, r]$ put a rectangle of height $f(r) = 1/r^2$. Now $f(x) > f(r)$ for $r-1 \leq x < r$, so this rectangle lies under the curve, and its area $1/r^2$ is strictly less than $\int_{r-1}^{r} f(x)\,\mathrm{d}x$. Adding for $r = n+1, n+2, \ldots$, whose intervals fill $[n, \infty)$,

$$\int_n^{\infty} \frac{\mathrm{d}x}{x^2} > \sum_{r=n+1}^{\infty} \frac{1}{r^2}.$$

(All three quantities are finite: the integral is $1/n$ and the sums are bounded by it.) This is the required pair of inequalities, and for the record they say

$$\sum_{r=n}^{\infty} \frac{1}{r^2} > \frac{1}{n} > \sum_{r=n+1}^{\infty} \frac{1}{r^2}. \qquad (*)$$

Trapping the tail

Write $S_n = \sum_{r=1}^{n} 1/r^2$. Then $A - S_n$ is exactly the tail $T = \sum_{r=n+1}^{\infty} 1/r^2$.

The right-hand inequality of $(*)$ gives $T < 1/n$ directly.

> **Ask yourself** The left-hand inequality of $(*)$ starts its sum at $r = n$, but the tail starts at $r = n+1$. How do we get a lower bound on T?

The inequality $(*)$ holds for every positive integer, so apply its left-hand half with $n+1$ in place of n:

$$T = \sum_{r=n+1}^{\infty} \frac{1}{r^2} > \int_{n+1}^{\infty} \frac{\mathrm{d}x}{x^2} = \frac{1}{n+1}.$$

Putting the two together,

$$\frac{1}{n} > A - \sum_{r=1}^{n} \frac{1}{r^2} > \frac{1}{n+1},$$

as required.

The smallest n

All terms are positive, so $S_n < A$ and the error is $A - S_n$.

- If $n \leq 9999$, then $n+1 \leq 10^4$, so $A - S_n > \dfrac{1}{n+1} \geq 10^{-4}$: the error is too big.
- If $n = 10^4$, then $A - S_n < \dfrac{1}{n} = 10^{-4}$: the error is small enough.

So the smallest value is $n = 10\,000$. Notice that the strictness of both inequalities matters here: at $n = 9999$ the lower bound is exactly 10^{-4}, and at $n = 10^4$ the upper bound is exactly 10^{-4}.

The improved estimate

Subtracting $\frac{1}{n+1}$ throughout the trapping inequality,

$$\frac{1}{n} - \frac{1}{n+1} > A - \left(\frac{1}{n+1} + \sum_{r=1}^{n} \frac{1}{r^2}\right) > 0.$$

So $\frac{1}{n+1} + S_n$ still underestimates A, by less than

$$\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}.$$

For $n = 10^4$ this is $\dfrac{1}{10^4 \times 10\,001} < \dfrac{1}{10^4 \times 10^4} = 10^{-8}$, so the improved estimate approximates A with an error of less than 10^{-8}, as required.

The extra term $1/(n+1)$ is a guess at the tail, and since the tail is pinned into an interval of width only $1/(n(n+1))$, the guess cannot be far wrong.

Checks

- Small case: with $n = 1$, $S_1 = 1$ and $A = \pi^2/6 \approx 1.6449$, so $A - S_1 \approx 0.645$, which does lie between $\frac{1}{2}$ and 1.
- Using $A = \pi^2/6$ and summing numerically: $A - S_{9999} \approx 1.00005 \times 10^{-4}$ (just too big) and $A - S_{10000} \approx 0.99995 \times 10^{-4}$ (just small enough), confirming $n = 10\,000$.
- For $n = 10^4$ the improved estimate is out by about 5.0×10^{-9}, below the bound $1/(n(n+1)) \approx 1.0 \times 10^{-8}$; it is roughly half the bound, as the picture suggests (the curve cuts each rectangle's overhang roughly in half).

Common slips

- Using the left-hand inequality at n rather than $n+1$, which gives a lower bound on $\sum_{r=n}^{\infty}$, not on the tail $A - S_n$.
- Answering $n = 9999$ or $n = 10\,001$ by treating a strict inequality as non-strict, or by using only one of the two bounds. You need the lower bound to rule out $n \leq 9999$ and the upper bound to accept $n = 10^4$.
- Drawing the rectangles on the wrong side of each sample point, which reverses the inequalities.

Beyond STEP

This is the integral test for convergence, together with its quantitative bonus: the same rectangles that prove a series converges also estimate its tail. Refining the idea, by comparing the sum with the integral and correcting with derivatives at the ends, leads to the Euler–Maclaurin formula, which here predicts $A - S_n \approx \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{6n^3}$ and was how Euler computed $\sum 1/r^2$ to many decimal places before he proved it equals $\pi^2/6$. Abbott's *Understanding Analysis* treats series and their convergence tests carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.